PLAN — Moon reading light

Status: **main board v0.1 + LED board v0.2 (18 mm, LP5012) and the slim moon head, through verification rounds 1–3, nothing ordered** (2026-11-03). The build pack is in `deliverable/` (start with `00-START-HERE.pdf`). Next: Matt orders boards + parts, prints the coupons, builds and bench-tests (docs/handoffs/, docs/verification/VERIFY.md). Single source of truth for requirements and decisions; reasoning is kept so later revisions don't re-derive it.

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1. Requirements

Statuses follow the 3dprint convention: **GIVEN** (Matt said it), **ASSUMED** (Claude's reading; confirm), **RULE** (engineering rule this design adopts).

#	Status	Requirement
R1	GIVEN	Clip-on reading light: a pack clips onto the back of a book with a slip-on clip
R2	GIVEN	Pack holds the battery and electronics; enclosed mini LiPo, USB-C charging
R3	GIVEN	From the top of the pack a flexible, adjustable neck ~10 cm long
R4	GIVEN	At the end of the neck a hollow model of the moon, ~5 cm diameter, lit fully from inside by LEDs
R5	GIVEN	One main button: press turns the light on/cycles colours (answer 3: the button never turns it off)
R6	GIVEN	Hold the main button > 2 s: sets the current colour as the power-on default
R7	GIVEN	Slide on/off switch on the side is the only on/off (answer 3)
R8	GIVEN	Up and down brightness buttons above and below the main button
R9	GIVEN	Colours: soft white, blue-white, yellow/orange/natural white, soft orange night light, red, green, blue; research and add other night-light whites
R10	GIVEN	Moon model generated from NASA data (answer 5), Thingiverse surveyed
R11	GIVEN	LED design: CRI-90+ tunable white + RGB with a constant-current driver (answer 1); LEDs in the base of the moon at the top of the neck
R12	GIVEN	Neck: pre-made metal, plastic-coated bendable arm
R13	GIVEN	Battery ~1200 mAh (answer 4)
R14	GIVEN	PCB files ready for PCBWay (or another fab); print files ready for the printer (Bambu X2D); BOM + order list for Farnell or RS
R15	GIVEN	Use the 14MM47/pcb and 14MM47/3dprint processes; product files live in this repo (answer 6)
R16	GIVEN	Multiple verification runs against datasheets, with adversarial/specialist reviewers
R17	RULE	Slide switch off = battery isolated from everything except the charger: standby drain < 10 µA
R18	RULE	LiPo charged CC/CV 4.20 V at ≤ 0.5 C; reverse-battery protected; load sharing so charging terminates with the light on
R19	RULE	Flicker-free (PWM ≥ 20 kHz) and no PLA moon softening: LED-board temperature limited in firmware with a sensor. LP5012: 29 kHz PWM; its 3-bit dither repeats at ~3.6 kHz with a 1/4096 step, well below visibility (VERIFY O5)
R20	RULE	Night modes contain no green or blue light (melanopic research); whites on the Planckian locus
R21	ASSUMED	Hand-assembly friendly parts (0603+, SOIC/TSSOP, no QFN/BGA) so all three assembly routes work. **Revised 2026-11-03 for the LED board v0.2** (Matt chose the LP5012): WQFN-20 + 0402 on its bottom face, fab assembly of that face recommended (`bom/assembly_routes.md`); the main board still meets R21
R22	ASSUMED	Pack ≤ ~45 × 80 × 20 mm, total mass ≤ ~100 g

2. Architecture

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 USB-C (power only, 5.1k CC) —PTC 1.1 A——TVS—┬—> MCP73831 charger 256 mA —> BAT —[Q3 rev-batt
P-FET]— LiPo 1200 mAh (PCM)
                                             └—> D3 S1J rectifier —┐
 BAT —[Q1 load-share P-FET, gate = VBUS, R5 10k]———————————————————┴—> VSYS_RAW —[Q2 P-FET,
slide switch]—> VSYS 3.3–4.2 V
 VSYS —> ATtiny3216 (buttons, I2C master, VDD measurement, PB2 → Q4)
 VSYS —[Q4 P-FET, MCU-switched]—> VLED (+ I2C pull-ups) — JST-SH 4-wire cable up the gooseneck
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   LED board v0.2 in the moon base (Ø18 mm, 1.0 mm): TI LP5012 12-ch constant-current sink, 25
mA/channel, 29 kHz
   ├— 4 × 2700 K + 4 × 5000 K Bridgelux Thrive 2835 (CRI 97), radial, ring r 5.8 mm     ├— 1 ×
Würth RGB at the centre
   └— MCP9800A0: firmware derating + ALERT thermostat → LP5012 EN (60/52 °C)             all
anodes on VLED
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- **Main board** (pack, 35 × 62 mm, 2-layer 1.6 mm, all parts top): see `hardware/main-board/design.py` docstring.
- **LED board v0.2** (moon base, Ø18 mm, 2-layer 1.0 mm, white mask, 0.15/0.15 mm rules; LEDs and the EN pull-up on top, driver/sensor bottom): `hardware/led-board/design.py`; reasoning in `docs/design/led-board-v0.2.md`.
- **Optics**: a Lambertian source on the wall of a sphere lights its inside uniformly (integrating-sphere geometry). The opening in the moon's bottom is the aperture, so the LED depth below it sets the evenness of the low wall. Model `docs/design/optics/head_v02.py` (output `head_v02.txt`): v0.2 head, opening r 9.0 (23.7°), emitters 1.2 mm below it: max/min **1.028** one white, **1.021** both whites, **1.037** RGB on the visible wall, with the RGB package shading the whites (round 3; v0.1 head: 1.068 / 1.063 / 1.054). The round-1 model `sphere_illum.py` is kept for the record. Raising the board above the opening gives ≥ 2:1.
- **Light output**: 4000 K full ≈ 37 lm out of the moon (61 lm of LED flux × ~0.61), **20–30 lux on a page at 25–30 cm**. That is a moon lamp / night-reading level, not a full reading light (round1 F6; next-revision option in docs/verification/round1.md §5).
- **Firmware**: `firmware/` — portable core (`lib/core`) + hardware glue (`src/main.cpp`); modes from `tools/gen_modes.py` (CIE 1931 mixing to the Planckian locus).
- **Mechanics**: lithophane moon (PLA, NASA LROC albedo + LOLA relief → 0.9–2.6 mm wall, relief inside). v0.2 head: the moon ends in a Ø25.3 mm skirt that slides over a Ø20.8 mm PETG cup (a slim stem, ~7 % of the sphere given up instead of ~13 %) and locks with an inside-out bayonet (3 lugs on the cup, grooves with a cam in the skirt, a 30° clockwise twist). A 0.8 mm PETG ring clamps the LED board, so the PLA never touches it; it rolls into a 0.4 mm gap above the cup's land, which makes the clamp a spring with a set preload (`LOCK_OFFSET`). The cup is jammed between two M10×1 nuts on the gooseneck's end. Pack = front shell + back plate + slip-on cantilever clip (PETG).
- **Thermal**: three layers. Firmware derating 45/52/58 °C; a 200 mA cap only with a trusted sensor (else 110 mA); the MCP9800 hardware thermostat on the LP5012's EN. Detail: docs/design/electronics.md §2.

3. Light modes (R9, R20)

Ten modes in button order (`firmware/include/modes_table.h`, generated): Warm 2700 K · Soft white 3000 K · Neutral reading 4000 K · Daylight 5000 K · Blue-white 6500 K · Candle 1900 K · Soft

orange night · Red night · Green · Blue. Evidence and targets: `docs/research/light-modes.md` (melanopic EDI limits from Brown et al. 2022, red for dark adaptation, Planckian chromaticities). 16 brightness steps, geometric 0.4 %–100 %. Power-on default: Soft white 3000 K at level 10 until the user saves another. With one RGB LED (v0.2) the red-based modes give about half the v0.1 flux (red night 1.1 lm, soft orange 1.5 lm, candle 2.4 lm) and 6500 K gives 30 lm instead of 40; Green and Blue are unchanged (one LED at full duty instead of two at half current) and dim gracefully below ~3.7 V of battery. Night modes go to level 15 (still < 1 lux at 30 cm).

4. Decision log

- 2026-11-02 — Product files in `reading_light`; process conventions from `14MM47/pcb` and `14MM47/3dprint` (Matt, answer 6).
- 2026-11-02 — LEDs in the moon base facing up (Matt's suggestion); confirmed by the optics model (uniform within ~1 %).
- 2026-11-02 — Slide switch is the only on/off; main button never turns off (Matt, answer 3).
- 2026-11-02 — Farnell UK chosen over RS (wider semiconductor range, Adafruit/TI/Microchip stock); RS only as fallback.
- 2026-11-02 — LED driver TLC59116 (Farnell 3119169) instead of TI LP5012 (not stocked by Farnell/RS). Cost: 25–37 mA driver supply current while on; 8-bit PWM (+ IREF gain for low-end resolution).
- 2026-11-02 — Cool white is 5000 K (only CRI-90+ cool 2835 at Farnell); 6500 K is mixed with blue + a little green.
- 2026-11-02 — RGB LED Würth 150141M173100 (datasheet + exact KiCad footprint) instead of Kingbright KAAF-5050 (datasheet unobtainable → pinout unverifiable). Würth abs max 30 mA → TLC59116 REXT 750 Ω = 25.0 mA/channel.
- 2026-11-02 — Two RGB LEDs (14 of 16 driver channels) to double red for candle/night modes.
- 2026-11-02 — USB→system OR-ing uses an S1J silicon rectifier, not a Schottky: MBR120VLSF reverse leakage (0.6 mA max @25 °C) would drain the cell through the load-share FET while switched off. Cost: VSYS ≈ 4.15 V on USB.
- 2026-11-02 — 10 µF capacitors are 0805 16 V (0603 10 V keeps only ~2.9 µF at 5 V per Samsung's DC-bias data).
- 2026-11-02 — LED board Ø26 mm (moon opening 33° instead of 30°): the 24 mm board could not be routed; optics model shows no uniformity change (max/min 1.001 at the chosen depth).
- 2026-11-02 — LM75BDP (TSSOP-8) instead of LM75BD (SOIC-8) to fit the LED board.
- 2026-11-02 — Moon relief on the inside, smooth 50.0 mm outside (crisp silhouette; outer overhangs ≤ 45° by claim).
- 2026-11-02 — Gooseneck: 120 mm, 10 mm OD, M10×1 ends, 6.8 mm bore (no coated gooseneck of this length is sold in the UK); sleeve it in silicone/heat-shrink for the plastic-coated look and feel (R12).

- 2026-11-02 — Verification round 1 (4 reviewers; docs/verification/round1.md). Changes adopted:
    - `board_map.h` is generated by `hardware/build.py`, with a consistency test (B1).
    - Q4 MCU-switched LED-board supply; pull-ups on VLED (M2, M-3).
    - J4 pin 1 = VREF via 1 k (M3).
    - Charge 256 mA, R3 3.9 k (M4).
    - F1 PTC ahead of the TVS (M-5, M-6).
    - R5 10 k (m1).
    - LM75B OS hardware thermostat on TLC RESET (M-1, m6).
    - Derating 45/52/58 °C, with the trusted-sensor 200/110 mA caps (F1, F4).
    - PETG clamp ring (F5).
    - Recipes regenerated for the Würth RGB at 25 mA (F7).
    - CG steps limited to the datasheet ranges (F9).
    - Green/Blue at half current (M1).
    - Firmware: WDT, bus recovery, read-back, timer-wrap fixes.
    - Pack: J2 relocated with a lead slot (B-3); 1.6 mm screw-head gap over the pouch (B-2).
    - LED board 1.0 mm in the board file (N9).
    - Router fixed to keep the whole track body out of rule areas.
- 2026-11-02 — Verification round 2 (2 reviewers; docs/verification/round2.md). Changes adopted:
    - Battery-lead slot 6.4 × 5.0 mm, so the pre-fitted plug passes.
    - Q4 soft turn-on (R13 10 k, C8 100 nF gate–drain).
    - F1 is now MF-MSMF110/16X; TP7 VLED test pad added.
    - LED-board RESET pull-up 2.4 k.
    - W/C silk letters at the white LEDs; values and DNP cross-outs on the assembly drawings.
    - Firmware: `cg_select()` moved into the host-tested core; faded cap changes; frozen timer from load start; SWRST and VLED bleed on power-cycle; LM75B read-back.
    - CAD: back-plate locators inside the cavity; 4.5 mm under the LED board; 0.88 mm board pocket; J2 headroom pocket; switch finger recess; larger USB-C window; button flange Ø9.0.
    - New `jig_led_board`; cross-part claims added.
- 2026-11-02 — `fab/*/v0.1/` committed before ordering (deviation from the pcb repo's "commit fab on order" rule): Matt asked for files he can take straight to the fab. The order reference and tag follow at order time (handoff 01).
- 2026-11-02 — Accepted for v0.1, revisit in v0.2: no cell NTC; reversed cell not blocked while USB is present; G/B tint drift at low battery; reading illuminance 20–30 lux. Decisions for Matt are listed in round1.md §5.
- 2026-11-03 — LED board v0.2 and a slim moon head (Matt: "go with LP5012, route the Ø18 board and fold in the full redesign"; aim: a smaller cap that takes less of the moon, driver kept on the LED board). Detail and reasoning: `docs/design/led-board-v0.2.md`.
    - TI LP5012 (WQFN-20, 3 × 3 mm, DigiKey UK: not at Farnell/RS) replaces the TLC59116: R1 2.94 kΩ = 25.0 mA, 0x14, 4–6 mA supply instead of up to 29 mA. R21 revised for this board (fab assembly of its bottom face recommended).

    - MCP9800A0 (SOT-23-5, Farnell) replaces the LM75BDP: the TSSOP-8 did not fit and the TMP1075's pin order forces a crossing; ALERT holds the LP5012's EN low (60/52 °C). The 2.4 kΩ EN pull-up moves to the board's top face.
    - 8 whites radial on a ring of radius 5.8 mm (dense 2835 courtyard footprint), one RGB at the centre (night/colour modes about half as bright; whites unchanged). Board Ø26 → Ø18 mm, 0402 passives.
    - 0.15 / 0.15 mm rules on the LED board (below the pcb repo's 0.2 mm working minimum, inside the fabs' standard rules): the WQFN's middle pins cannot escape at 0.2 mm.
    - Cathode fan-in and the I²C/EN bundle are fixed copper in design.py; kigen gained grid alignment, `skip_connected` stitching and optional (pour-finished) nets. The main board still routes clean with them (rebuild checked; its committed files are unchanged).
    - `fab/led-board/v0.1/` removed (never ordered); `fab/led-board/v0.2/` written.
    - Moon head: inside-stem cup Ø20.8 / skirt Ø25.3 (was a Ø38.7 cup), inside-out bayonet with a 0.25 mm cam, clamp ring 0.8 mm with a 45° chamfer under the moon's clamp cone, opening r 9.0, LEDs 1.2 mm below it.
    - **Correction**: an earlier sweep this session reported ~1.01 uniformity for a 30° flared bore with the LEDs 2.9 mm deep, but its ray test let light through the PLA where the flare was wider than the opening. Remodelled with the opening as the aperture (`docs/design/optics/head_v02.py`): depth matters; 1.2 mm chosen (1.011 / 1.002 / 1.037), and moon_shell B14 now checks that nothing solid sits in the emitter-to-opening cone.
    - CAD pipeline: wall-thickness sampling skipped zero-area faces wrongly and crashed; fixed with a unit test (`cad/tests/test_pipeline_checks.py`). No check or threshold was relaxed.
    - LED board C2 is 10 µF 0603 10 V (an exception to the 0805 16 V rule above: no room on the 18 mm board). At VLED ≤ 4.2 V it keeps roughly half its value, which is enough as local bulk; the LED board now totals ~6 µF effective (v0.1 ~9 µF), so Q4's soft turn-on sees less inrush.
    - Green/Blue can no longer run at a 12.5 mA step (the LP5012 has none): they run at 25 mA peak and fall out of regulation below ~3.7 V of battery. Accepted; option R1 3.65 kΩ (20 mA on all channels). VERIFY E9.

- 2026-11-04 — Verification round 3 (3 reviewers on LED board v0.2, its firmware and the slim head; docs/verification/round3.md). Changes adopted:

    - Clamp: the locked bayonet was line-to-line, so the fit depended on print effects. The cup's land now sits 0.4 mm under the board top (the ring rolls into the gap: a spring), and the locked groove floor stands `LOCK_OFFSET` 0.15 mm above the lug: 0.05–0.25 mm preload inside 0.23–0.39 mm of travel for a 0.9–1.1 mm board, checked by claims. `GAP_45` 0.35.
    - The moon locks clockwise seen from the top (it tightened nothing before and loosened the thread); the cup is jammed between two nuts (the gooseneck end has no shoulder); BOM 2 nuts + a washer per lamp.
    - Bayonet coupon split into a PETG cup half and a PLA skirt half (the moon's material and process), discs 0.9 / 1.0 / 1.1 mm. Ring ID 16.4, printed at 0.12 mm. 0.3 mm skirt lead-in.
    - CAD pipeline: per-facet wall samples added; area weighting alone had passed a 0.03 mm lip at the moon's groove roofs and a fold at its north pole. Both fixed in the geometry (roofs below the cone line; pole relief faded to the ring mean), with a unit test for the check.

    - Optics model: RGB package as an occluder and slant-corrected wall area: 1.028 / 1.021 / 1.037, escape 0.61.
    - LED board: IREF as fixed copper from the pad centres (a 0.106 mm copper neck at the routed joint) and a `connection_width` DRC rule. Accepted for this revision: VCAP/RIREF ground returns of ~22–26 mm round the I²C bundle (DC error ~0.2 %; VERIFY E18 scopes VCAP; EP vias + a GND jumper next revision).
    - Firmware: outputs zeroed in standby before Chip_EN; CONFIG and output registers read back every second; the thermostat hold no longer counts towards a power-cycle; the first reading after init sets the derate at once; a sensor that has risen 1 °C under load keeps its trust at equilibrium (no "breathing"); 32 host tests.
    - Green/blue headroom stated for the worst die (~4.05–4.15 V needed); VERIFY E9 measures the received dies first.

- 2026-11-08 — SPICE simulation of both boards (`sim/`, report rev 1 `docs/verification/spice-report_main-v0.1_led-v0.2_rev1.md` and `.pdf`): 33 runs, 35 checks. Nothing changed in the design. Findings for Matt: the USB-unplug handover dips VSYS 0.64 V for ~21 ms (Q1 body diode; E12's R5 remedy only shortens it); hot-plugging J3 at a low battery can reset the MCU (E13); VSYS on a 4.75 V source is 3.71 V (E15 assumes 5.0 V).

- 2026-11-08 — SPICE decisions (Matt; report rev 2 `docs/verification/spice-report_main-v0.1_led-v0.2_rev2.md`, rev 1 kept as `_rev1`):

    - D2 accepted with conditions: J3 is mated once at assembly with SW1 OFF and strain-relieved (handoff 05 §4.3); the hot-plug step is removed from VERIFY E13.
    - D3 accepted: a 5 V ≥ 1 A USB-C charger is specified (handoff 05 user notes) and E15 runs on 5.0 V; lowering `VBAT_RESUME_MV` to 3600 only if E15 shows a late wake; VBUS sense for the next revision.
    - D4 accepted: whites added to the E9 V_F measurement; raise `VBAT_LOW_MV` to 3500 only if a white reads > ~2.85 V.
    - D5: keep I²C at 100 kHz. D6: Q4 soft-start current note corrected in electronics.md §1 (25–36 mA simulated).
    - D1: next-revision study done (parts survey + simulation). Recommended: TI LM66100 in place of Q1, CE straight to VBUS, S1J kept, R5 1k (dip < 1 ms; +≤ 0.3 µA standby). A VBUS detector driving Q1 (or any bidirectional switch at a raised threshold) is unsafe: USB can charge the cell past the charger. v0.1 handling of E12 still to confirm.

- Next main-board revision (from the SPICE study): LM66100 battery-path ideal diode (D1), VBUS sense to an MCU pin (D3); check LM66100 stock (none at DigiKey UK until 2026-12-21). Alternative: TPS2116 power mux, only after re-checking LED-board thermal design and the LP5012 5.5 V margin (VSYS ≈ VBUS on USB).

5. Open questions / VERIFY

See `docs/verification/VERIFY.md` (measurements that only the prototype can settle).

6. Status

Phase	Deliverable	Status
Requirements & research	this file, docs/research	done

Phase	Deliverable	Status
Electronics	main board v0.1 and LED board v0.2: ERC 0 / DRC 0 errors / 0 unconnected / 0 parity; fab/main-board/v0.1, fab/led-board/v0.2	done
Firmware	firmware/ (LP5012 + MCP9800 layer), host tests + AVR builds	done (v0.2 LED board), untested on hardware
Mechanics	cad/parts (11 parts PASS, v0.2 head), print/ STL + 3MF	done
Verification rounds	docs/verification/round1.md, round2.md, round3.md, VERIFY.md	rounds 1–3 done
SPICE simulation	sim/ (ngspice), docs/verification/spice-report_main-v0.1_led-v0.2_rev2.md and .pdf (rev 1 kept)	rev 2 2026-11-08: D2-D6 decided and applied; D1 next-revision part chosen, v0.1 E12 handling open
BOM/order, handoffs	bom/ (Farnell + DigiKey), docs/handoffs 01–06	done
Build pack	deliverable/ (generated by tools/build_deliverable.py)	done